Characterizing host-virus interactions through host factor curation and network analysis

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INTRODUCTION

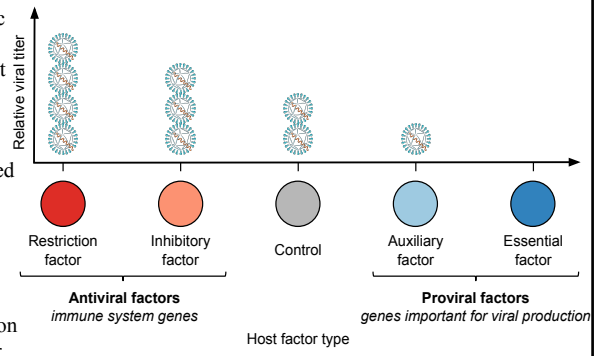
- The biomolecular interactions that take place between a virus and its host are complex in nature and play a significant role in governing functional and evolutionary dynamics.
- The mechanics by which viruses infect, replicate, and otherwise impact its host varies across individuals, species, and between generations.
- Determining what interactions occur, how they evolve, and their importance in the context of the cellular interactome, or the set of interactions between genes within a cell, is of significant value and has yet to be fully established.
- This gap in knowledge makes the development of an effective treatment against viral infection difficult and also hinders the establishment of efficient cell-based vaccine manufacturing platforms.

OBJECTIVES

1. Curate and annotate anti- & pro-viral genes, collectively known as host factors, that have evidence of interacting with the influenza virus.
2. Build a reference host-virus interactome and explore its network properties using the literature-derived host factors from (1) as a guide.
3. Compare results from (2) with a putative antiviral gene set derived from a genome-wide knockout screen of cells infected with influenza. This will generate a set of candidate antiviral genes that may be engineered to optimize cell-based vaccine production.

METHODS

Figure 1: Impact of genetic perturbation on the viral production capacity of host cells. 46 host factors with experimental evidence of interacting with the influenza virus were curated from the literature and subsequently classified based on the scheme described on the right. Shown is the relative viral titer produced by a cell upon perturbation of a particular type of host factor.



METHODS (continued)

- Host factor pairs (interactions) were plotted in gene coexpression space based on data obtained from COXPRESdb[1]. Each gene was annotated as either anti- or pro-viral, as depicted in Figure 1.
- A ranked list of putative antiviral genes were derived computationally through analysis of next-generation sequencing data from a novel genome-wide knockout screen in HEK293 cells infected with influenza[2].

RESULTS

- Gene coexpression data, as measured between pairs of genes, can be used as a proxy for functional interactions.
- The bimodal distribution of gene coexpression scores for antiviral gene pairs, as observed in the red violin plot in Figure 2, suggests that antiviral host factors act in one of two ways:
  1. As part of the host cells' non-specific innate immune system, where coexpression between genes within this pathway is enriched upon invasion by a pathogen (peak at ~2000).
  2. As direct, specific inhibitors of the influenza virus, where coexpression with other antiviral factors is limited (peak at dotted line).

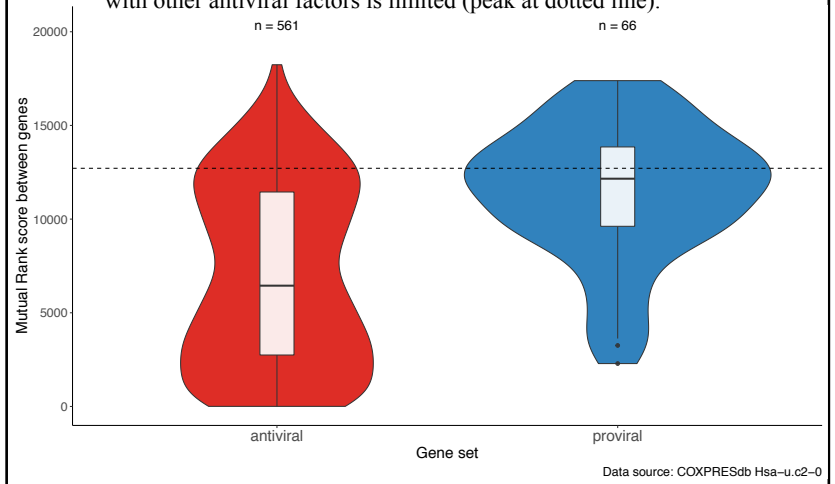


Figure 2: Gene coexpression distributions stratified by type of host factor. Each point within a distribution is the Mutual Rank[1] score between a pair of genes, where a small value denotes high coexpression and vice versa. The dotted horizontal line denotes the average gene coexpression score for all genes within the human genome.

RESULTS (continued)

- Antiviral gene hits derived from perturbation-based screens that are specific to the influenza virus have weak, if any, enrichment between each other.
- Fold enrichments with the new Sharon *et al.* (2020) dataset are 1.1 and 1.9 for Heaton and Tripathi, respectively. Fold enrichment between Heaton and Tripathi is 0.6.

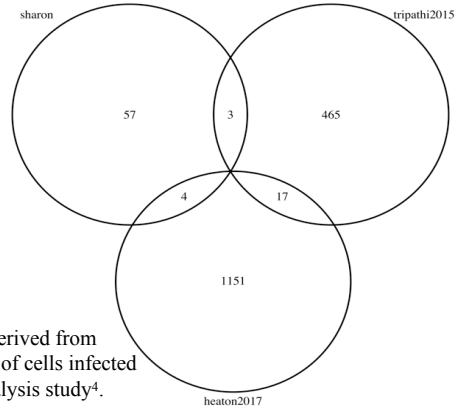


Figure 3: Overlap of antiviral gene sets derived from perturbation-based genome-wide screens of cells infected with the influenza virus[2,3] and a meta-analysis study[4].

CONCLUSIONS & FUTURE WORK

- Functional interactions between host factors that impact viral production capacity in human cells can be determined by proxy through analysis of gene coexpression data.
- A robust set of antiviral genes derived from perturbation-based screening experiments and network analyses may be used to genetically engineer host cells to increase viral output for use in cell-based vaccine manufacturing platforms.
  - This is of critical importance in pandemic situations as it has the potential for rapid scale up and a concomitant reduction in vaccine production timelines.
- Further insight into the interactions that govern influenza virus infection can be obtained through analysis of other types of interactome data, such as physical protein-protein interactions and tissue-specific viral-induced transcriptomics data.

REFERENCES & ACKNOWLEDGEMENTS

1. T. Obayashi *et al.*, Nucleic Acids Research 2019.
2. D. M. Sharon, S. Nesdoly *et al.*, accepted 2020.
3. B. E. Heaton *et al.*, Cell Reports 2017.
4. S. Tripathi *et al.*, Cell Host & Microbe 2015.

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